

Math 1450 (Algebra and Trigonometry) Section 02 Syllabus—Fall 2019

Instructor: Professor Bryce Christopherson

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Office Hours: MTWF 10:00 AM–10:50 AM
in RH 315

Class: MTWRF 9:00 AM–9:50 AM in AG 4041

Course Description: From the University Course Catalog: Emphasizes aspects of algebra, trigonometry and problem solving important in the study of calculus. Functions and their applications to real world problems. Classes of functions including polynomial, exponential, logarithmic and trigonometric functions. Intuitive introduction to the idea of limit and sequence, which are developed further in the calculus sequence. For the student with considerable prior exposure to trigonometry and algebra. See instructor for specifications. Students with both MATH 1400 and 1405 credit may not receive credit for this course.

Prerequisites: You must satisfy one of the following requirements to be enrolled in this course:

- A placement level III from the math placement exam
- An ACT Math score 23, no more than 3 years old
- Passing MATH 925 (C or better), or
- Earning D or better in either Math 1400 or Math 1405.

Students are expected to have learned basic trigonometry before entering Math 1450. Others should enroll in the sequence Math 1400/1405. You must earn a grade of C or better to enroll in Math 2200.

Course Materials: The materials for this course, both required and optional, are as follows:

Course Textbook (Required): The required text for this course is Precalculus, an OpenStax resource. This textbook was written to increase student access to high-quality learning materials, while maintaining a high standard of academic rigor at little to no cost. Various versions of this textbook can be downloaded or ordered through the website: <https://www.openstax.org/details/books/precalculus>.

Software (required): We will use the XYZHomework online homework system in this course. Details on how to enroll in the XYZHomework online homework system will be discussed on the first day of class.

**For best results, register for XYZHomework and
start working on your homework the first day of class.**

Internet access (required): You will need internet access to use XYZHomework to finish your online homework.

Calculator (required): A scientific non-graphing calculator is recommended for this course. During exams and quizzes, *you will not be allowed to use a graphing calculator or a cell phone* and you will not be allowed to share a calculator with someone else.

Note Cards (optional): For each exam you will be allowed to use both sides of one 3×5 index card for hand-written notes.

WyoCourses: All materials for this course will be posted the the class WyoCourses page. Any updates or changes pertaining to the course will only be posted as an announcement in WyoCourses. Adjust your notification settings so that you receive emails or texts when announcements are posted. You want to be sure that you do not miss updates on assignments or other important information.

Grading: The grade you receive at the end of this course will be based on your performance in several categories, which determine your final grade percent according to the table on the left. The approximate grading scale used to evaluate this total percentage is given in the table on the right.

Assignment Group	Grade Percentage
Exams	80%
... Exam 1	20%
... Exam 2	20%
... Exam 3	20%
... Final Exam	20%
XYZHomework	20%
Total	100%

Letter Grade	Overall Percentage
A	[90,∞)
B	[80,90)
C	[70,80)
D	[60,70)
F	[0,60)

Exams (80%): Exams 1-5 and the Final Exam will be given outside the regularly scheduled class time on the dates shown below. Locations will be announced later during the semester.

Exam No.	Date	Location	Sections Covered
Exam 1	Thurs, Feb. 27, 5:00 PM–7:00 PM	CR 105	1.1-1.7, 2.1-2.3, 3.1-3.8
Exam 2	Thurs, Mar. 26, 5:00 PM–7:00 PM	CR 105	4.1-4.6, 5.1-5.4
Exam 3	Thurs, Apr. 16, 5:00 PM–7:00 PM	CR 105	6.1-6.3, 7.1-7.5, 8.1-8.5
Final Exam	Thurs, May. 14, 3:30 PM–5:30 PM	TBA	9.1-9.3, 11.1-11.4, 12.1-12.4

Make-up Exams: If you will not be able to take an exam on the scheduled date and time you must make other arrangements with the instructor well in advance. In cases where proper notification has not been given, or if the reason for missing the exam is not valid, the make-up the exam may be penalized or may not be given.

Online Homework (20%): The instructions for registering for XYZHomework can be found on the WyoCourse site. Online assignments should usually take around 30 minutes to complete, on average. These problems are chosen as representative of the basic concepts presented during lecture.

For full credit, XYZHomework assignments must be completed by 11:59pm on their due date. You have an limit of 3 attempts on each problem. None of the XYZHomework assignments will be dropped. Check the attached course schedule or the "Assignments" tab on the class WyoCourses page for the assignment due dates.

Late Passes: In XYZHomework, you have the option to use a "Late Pass" to extend the submission deadline for an assignment by 24 hours. This extension is applied to the *original* due date, *not* the current date. So, for example, if you have an assignment that is 7 days past due and you would like an additional 24 hours to complete it, you would need to use 8 Late Passes to do so. By default, you will begin the class with one Late Pass for each assignment (for a total of 52 late passes). Except in exceptional circumstances, I will not grant more Late Passes beyond this.

Think of using Late Passes like this: You begin the class with as many Late Passes as there are assignments. This means that you have about a day of 'wiggle room' on each assignment's due date. That is, you have enough Late Passes to do every assignment a day late. For every assignment you finish *two* days late, you'll need to finish *one* assignment on time (since using two Late Passes on one assignment means that you can't use the second Late Pass on another assignment). Things continue on in this pattern: for every assignment you finish *three* days late, you'll need to finish *two* assignments on time, and so on.

In summary, every time you use Late Passes to do your homework late, you *lose* a bit of leeway on deadlines for future assignments. Conversely, every time you do your homework on time, you *earn* a bit more leeway on deadlines for future assignments.

Extra Credit: Opportunities for earning extra credit may occur throughout the semester at the discretion of the instructor. Extra credit problems may be given in class (Note: such problems will not be emailed to anyone not in class at the time) or occasionally may be offered on quizzes and exams. Points earned from extra credit may help borderline grades, but do not expect extra credit to significantly change your final score.

Email correspondence: Please always put “Math 1450” in the subject line of any email you send to me. This will help me spot your email more easily, which translates to a quicker response. An email should have a salutation, body, and signature.

Attendance: Attendance to regularly scheduled lecture sessions is not mandatory, but I believe it is very important. If you frequently do not attend lecture without an exceptionally good reason and begin to struggle in the course, I will have very little sympathy for you.

Getting Help: One of the most important skills a student can learn is recognizing early that they are confused with some concept or skill in the class. Once you recognize you need help, first check through your materials, such as class notes, to see if the answer is there. If you cannot solve it yourself, *please seek help*. There are lots of resources available to you to help you succeed!

Study Groups: Meet with some of your classmates and help each other with questions. Learning from your peers is one of the best ways to learn.

Office Hours: My office hours (subject to change as noted above) are 10:00 AM - 10:50 AM, MTWF in RH 315. Please seek me outside of class if at all possible; and if these times are not convenient, we should be able to arrange an alternative time. I am here to help you.

The Center for Assistance in Statistics and Mathematics (CASM): The CASM is a free drop-in tutoring center for students enrolled in 1000 and 2000 level math courses. It is located in Ross Hall 29 (northwest corner, on bottom floor). They are open Monday-Thursday 10:00 AM–5:00 PM and Friday 10:00 AM–1:00 PM. It's also a great place to study, work on homework, or meet with study groups. They have copies of current textbooks for students to borrow while there as well as several desktop computers, a printer, and two study rooms.

Supplemental Instruction(SI): SI is a series of out-of-class study sessions led by a student who has successfully completed the calculus sequence. Meeting times and location will be announced by email early during the semester.

STEP Tutoring: The STEP Tutoring Program (<http://www.uwyo.edu/step>) offers drop-in one-on-one tutoring (no appointments) for Calculus students in Coe Library. Hours are Sunday-Thursday 5–9pm. Check in at the Research Help Desk on the main level. Online tutoring (eTutoring) is also offered. For more information visit <http://www.uwyo.edu/studentaff/step/tutoring/>

Walk-In Academic Coaching The Center for Advising and Career Services in 222 Knight Hall offers walk-in coaching to any student. They can help on a variety of topics, such as Note Taking, Time Management, Exam Preparation, Textbook Reading, Staying Motivated, Managing Finals, etc.

Student Learning Center in Washakie: The Student Learning Center (SLC), located in the lower level of the Washakie Center, offers free academic support services to students on a drop-in basis Sunday through Thursday. Please see <http://www.uwyo.edu/reslife-dining/slc/> for more information.

It is important that you try a variety of different services. If you do not receive the help you need at that time, please try it again or try a different service. Don't be shy—make sure you get the help you need!

Academic Dishonesty, Classroom Conduct, Disability Student Services, etc.: Please refer to the links provided through the class WyoCourses page.

Changes to Syllabus: All deadlines, requirements, policies, and course structure is subject to change if deemed necessary by the instructor. Students will be notified of minor changes verbally in class and substantive changes shall be communicated in writing distributed either by an announcement posted to our class WyoCourses page or via email.

Course Outline: The following table outlines the topics and relevant sections of the textbook that will be covered in this course. This is an overview, a more precise schedule can be found on the next page.

Topic	Sections	Topic	Sections
Functions	1.1-1.7	Trigonometric Identities and Equations	7.1-7.5
Linear Functions	2.1-2.3	Further Applications of Trigonometry	8.1-8.7
Polynomial and Rational Functions	3.1-3.8	Systems of Equations and Inequalities	9.1-9.4
Exponential and Logarithmic Functions	4.1-4.6	Sequences and Series	11.1-11.4
Trigonometric Functions	5.1-5.4	Introduction to Calculus	12.1-12.4
Periodic Functions	6.1-6.3		

Grading Rubric: The rubric below will be used *as a guideline* when evaluating your written work on homework assignments and exams throughout this course. You should strive to produce written solutions that are clear and well organized, make proper use of mathematical notation and terminology, demonstrate an understanding of the mathematical concepts involved, and result in a correct solution. If you practice these skills while working on your written homework problems, you will be more likely to meet the standard expected of your written work on exams. Please familiarize yourself with this rubric and reference it if ever you feel your work has been unfairly graded.

Characteristics of the Written Work	Qualitative Description	Typical Scoring %
(i) The work is presented in a neat, clear, organized fashion that is complete and easy to follow. (ii) Graphs are correct, complete, carefully drawn and appropriately labeled. (iii) Correct terminology and notation are always used. (iv) The solution shows a complete understanding of the mathematical concepts involved, although it may contain a minor arithmetic error or sign mistake.	Exceptional	90-100
(i) The work may not be completely clear or well organized, or it may have some steps missing, but it is still possible to follow with little effort. (ii) Graphs are correct and complete, but somewhat carelessly drawn or missing some labels. (iii) Correct terminology and notation are usually used. (iv) The solution shows almost complete understanding of the mathematical concepts and may contain one or two arithmetic or sign errors.	Good	75-90
(i) The work may be unclear, poorly organized, or have several steps missing, but it is possible to deduce the intended method of solution with some effort. (ii) Graphs are partially correct, somewhat carelessly drawn, and may be missing some labels. (iii) Correct terminology and notation are sometimes used. (iv) The solution shows a limited understanding of the mathematical concepts involved, or an almost complete understanding of the mathematical concepts involved, but contains several arithmetic or sign errors.	Fair	60-75
(i) The work is poorly organized or is very incomplete, making it difficult to follow. (ii) Graphs are incorrect, carelessly drawn, or incomplete, and may not be labeled. (iii) Correct terminology and notation are seldom used. (iv) The solution shows a limited understanding of the mathematical concepts involved, and may also contain several arithmetic or sign errors.	Poor	40-60
(i) The work is so disorganized or incomplete that it is very difficult or impossible to follow. (ii) Graphs are not drawn or are completely wrong and not labeled. (iii) There is little or no correct usage of terminology and notation. (iv) The solution shows little or no understanding of the mathematical concepts involved.	Very Poor	0-40

Course Schedule:

Sections Covered

1.1-1.7, 2.1-2.3, 3.1-3.8, 4.1-4.6, 5.1-5.4, 6.1-6.3, 7.1-7.5, 8.1-8.3, 8.5, 9.1-9.3, 11.1-11.4, 12.1-12.4

Su.	Mo.	Tu.	We.	Th	Fr.	Sa.
26 Jan.	27 Jan.	28 Jan.	29 Jan.	30 Jan.	31 Jan.	1 Feb.
	Intro	The Basics	1.1 <u>HW-Intro</u>	1.2 <u>HW-1.1</u>	1.3 <u>HW-1.2</u>	
2 Feb.	3 Feb.	4 Feb.	5 Feb.	6 Feb.	7 Feb.	8 Feb.
	1.4 <u>HW-1.3</u>	1.5 <u>HW-1.4</u>	1.6 <u>HW-1.5</u>	1.7 <u>HW-1.6</u>	2.1 <u>HW-1.7</u>	
9 Feb.	10 Feb.	11 Feb.	12 Feb.	13 Feb.	14 Feb.	15 Feb.
	2.2 (Graphing) <u>HW-2.1</u>	2.3 <u>HW-2.2</u>	3.1 <u>HW-2.3</u>	3.2 <u>HW-3.1</u>	3.3 <u>HW-3.2</u>	
16 Feb.	17 Feb.	18 Feb.	19 Feb.	20 Feb.	21 Feb.	22 Feb.
	3.4 (Graphing) <u>HW-3.3</u>	3.5 <u>HW-3.4</u>	3.5	3.6 <u>HW-3.5</u>	3.7 <u>HW-3.6</u>	
23 Feb.	24 Feb.	25 Feb.	26 Feb.	27 Feb.	28 Feb.	29 Feb.
	3.7/3.8	3.8 <u>HW-3.7</u>	Review <u>HW-3.8</u>	Review Exam 1 (Ch.1,2,3)	4.1	
1 Mar.	2 Mar.	3 Mar.	4 Mar.	5 Mar.	6 Mar.	7 Mar.
	4.2 <u>HW-4.1</u>	4.3 <u>HW-4.2</u>	4.4 (Graphing) <u>HW-4.3</u>	4.5 <u>HW-4.4</u>	4.6	
8 Mar.	9 Mar.	10 Mar.	11 Mar.	12 Mar.	13 Mar.	14 Mar.
	4.6 <u>HW-4.5</u>	5.1 <u>HW-4.6</u>	5.2 <u>HW-5.1</u>	5.2	5.3 <u>HW-5.2</u>	
15 Mar.	16 Mar.	17 Mar.	18 Mar.	19 Mar.	20 Mar.	21 Mar.
	Spring Break	Spring Break	Spring Break	Spring Break	Spring Break	

Su.	Mo.	Tu.	We.	Th	Fr.	Sa.
22 Mar.	23 Mar.	24 Mar.	25 Mar.	26 Mar.	27 Mar.	28 Mar.
	5.3/5.4	5.4 <u>HW-5.3</u>	Review <u>HW-5.4</u>	Review Exam 2 (Ch.4,5)	6.1/6.2 (Graphing)	
29 Mar.	30 Mar.	31 Mar.	1 Apr.	2 Apr.	3 Apr.	4 Apr.
	6.3 <u>HW-6.1</u>	7.1 <u>HW-6.2</u>	7.2 <u>HW-6.3</u>	7.3 <u>HW-7.1</u>	7.4 <u>HW-7.2</u>	
5 Apr.	6 Apr.	7 Apr.	8 Apr.	9 Apr.	10 Apr.	11 Apr.
	7.5 <u>HW-7.3</u>	7.5 <u>HW-7.4</u>	8.1 <u>HW-7.5</u>	8.2 <u>HW-8.1</u>	8.3 <u>HW-8.2</u>	
12 Apr.	13 Apr.	14 Apr.	15 Apr.	16 Apr.	17 Apr.	18 Apr.
	8.5 <u>HW-8.3</u>	8.5	Review <u>HW-8.5</u>	Review Exam 3 (Ch.6,7,8)	9.1	
19 Apr.	20 Apr.	21 Apr.	22 Apr.	23 Apr.	24 Apr.	25 Apr.
	9.1	9.2 <u>HW-9.1</u>	9.2	9.3 <u>HW-9.2</u>	11.1 <u>HW-9.3</u>	
26 Apr.	27 Apr.	28 Apr.	29 Apr.	30 Apr.	1 May.	2 May.
	11.2 <u>HW-11.1</u>	11.3 <u>HW-11.2</u>	11.4 <u>HW-11.3</u>	12.1 <u>HW-11.4</u>	12.2 <u>HW-12.1</u>	
3 May.	4 May.	5 May.	6 May.	7 May.	8 May.	9 May.
	12.3 <u>HW-12.2</u>	12.4 <u>HW-12.3</u>	12.4	Review <u>HW-12.4</u>	Review (Last day)	
10 May.	11 May.	12 May.	13 May.	14 May.	15 May.	16 May.
	Finals Week	Finals Week	Finals Week	Final Exam (Ch. 9,11,12)	Finals Week	